

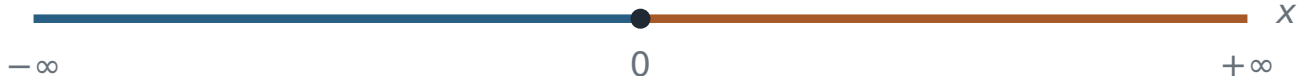
Left H^1 solution

$$\nabla\psi(0^-) = Tv$$

Right H^1 solution

$$\nabla\psi(0^+) = -Sv$$

$$\psi(0) = v$$



$$\nabla\psi(0^+) - \nabla\psi(0^-) = -(S + T)v$$



$$(-\nabla^2 + V + tI)\psi = \delta_0(S + T)v$$

Fourier inversion at $x = 0$, for every v

$$v = \frac{1}{2}R(t)(S + T)v \implies L = \frac{S+T}{2} = R(t)^{-1}$$